

Decision Trees and their variants

Naeemullah Khan

naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية

King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

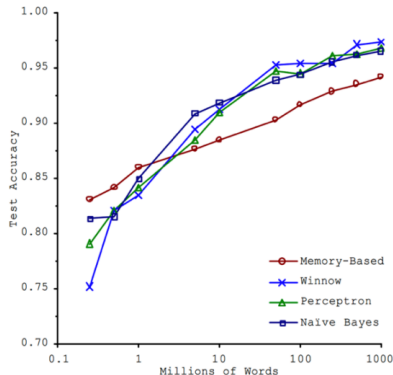
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► Brawn or Brains?

- In 2001, Microsoft researchers ran a test to evaluate 4 different approaches to ML-based language translation

► Findings:

- Size of the dataset used to train the model mattered more than the model itself
- As the dataset grew large, performance difference between the models became small



Banko, M. and Brill, E. (2001), "Scaling to Very Large Corpora for Natural Language Disambiguation"

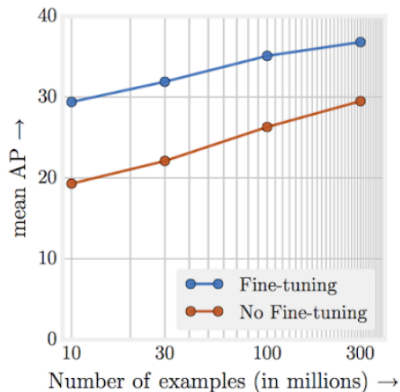
► The Unreasonable Effectiveness of Data

- In 2017, Google revisited the same type of experiment with the latest Deep Learning models in computer vision

► Findings:

- Performance increases logarithmically based on volume of training data
- Complexity of modern ML models (i.e., deep neural nets) allows for even further performance gains

► Large datasets + large ML models = amazing results!!



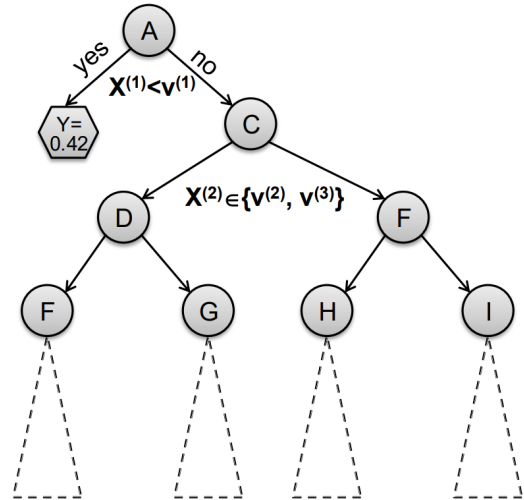
“Revisiting Unreasonable Effectiveness of Data in Deep Learning Era”: <https://arxiv.org/abs/1707.02968>

- ▶ A few reasons why this is important:
- ▶ They outperform DL models in certain tasks.
- ▶ Deep models are often hard to scale and require lots of data. Traditional models allow you to encode prior knowledge better and give you more control.
- ▶ Combine: ideas from several ML models, e.g., GNNs
- ▶ Rule of thumb: If working on a well understood problem use deep learning. If working on a new problem use techniques we'll discuss here.

- ▶ Decision trees are part of ML since 1980s
 - Introduced by Leo Breiman in 1984
 - Notable algorithms: ID3, C4.5
- ▶ More recent innovations include:
 - Boosted decision trees (gradient boosted DT)
 - Random forest
- ▶ Even though DTs are old, hand-engineered and heuristic, they are a method of choice for tabular data and for Kaggle competitions.

- ▶ Given one attribute (e.g., lifespan), try to predict the value of new people's lifespans by a subset of the other available attributes
- ▶ Input attributes:
 - d features/attributes: $x^{(1)}, x^{(2)}, \dots, x^{(d)}$
 - Each $x^{(j)}$ has domain O_j
 - ▶ Categorical: $O_j = \{male, female, nonbinary\}$
 - ▶ Numerical: $H_j = (1, 200)$
 - Y is output variable with domain O_Y :
 - ▶ Categorical: Classification e.g. $Y = \text{eye color}$
 - ▶ Numerical: Regression e.g. $Y = \text{lifespan}$
- ▶ Data D :
 - n examples (x_i, y_i) where x_i is a d -dim feature vector, $y_i \in O_Y$ is output variable
- ▶ Task:

- ▶ A Decision Tree is a tree-structured plan of a set of attributes to test in order to predict the output

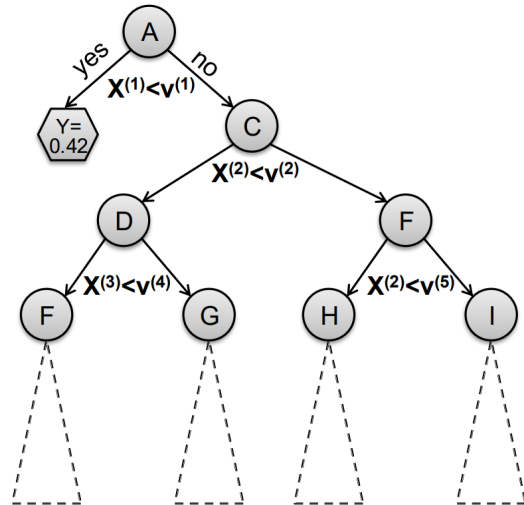


► **Decision trees:**

- Split the data at each internal node
- Each leaf node makes a prediction

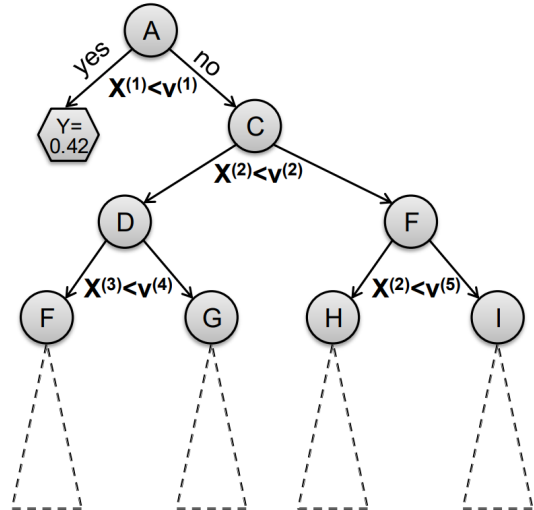
► **Lecture today:**

- Binary splits: $x^{(j)} < v$
- Numerical attributes
- Regression



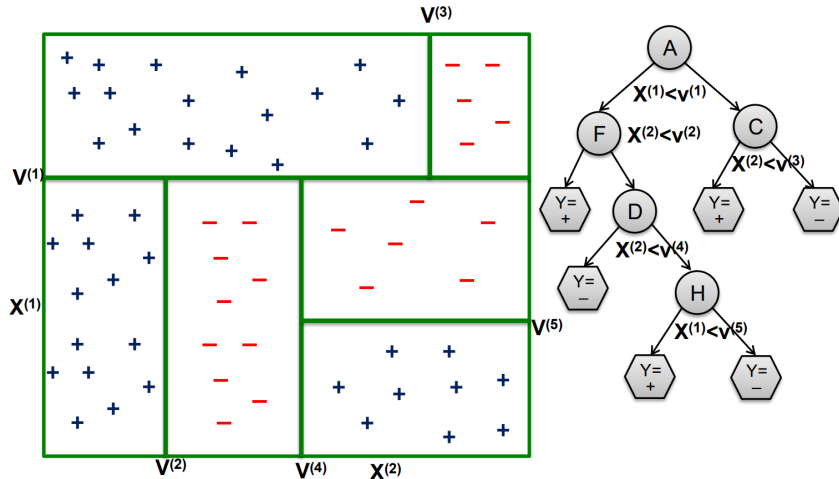
How to make predictions?

- ▶ Input: Example x_i
- ▶ Output: Predicted $\hat{y}_i$
- ▶ “Drop” x_i down the tree until it hits a leaf node
- ▶ Predict the value stored in the leaf that x_i hits



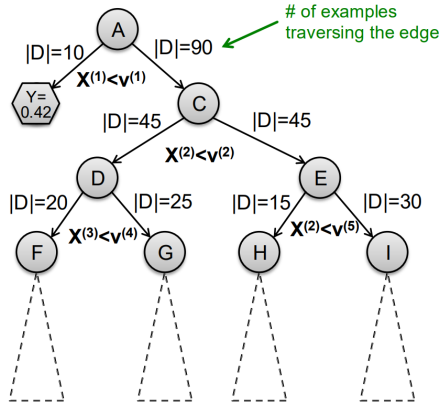
► **Alternative view:**

Decision Trees: feature space (cont.)



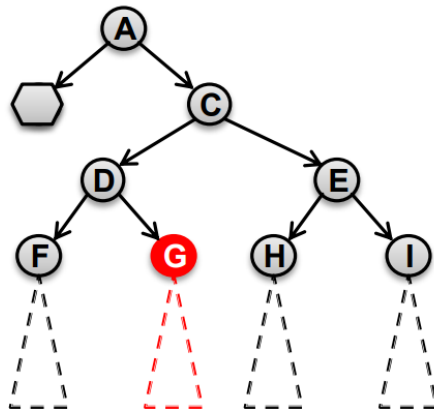
Training dataset D^* , $|D^*| = 100$ examples

- ▶ Training dataset D^* , $|D^*| = 100$ examples



How to construct a tree?

- ▶ Imagine we are currently at some node G
 - Let D_G be the data that reaches G
- ▶ There is a decision we have to make: Do we continue building the tree?
 - If yes, which variable and which value do we use for a split?
 - ▶ Continue building the tree recursively
 - If not, how do we make a prediction?
 - ▶ We need to build a “predictor node”



Algorithm 1 **BuildSubtree**

Require: Node n , Data $D \subseteq D^*$

1: $(n \rightarrow \text{split}, D_L, D_R) = \text{FindBestSplit}(D)$ (1)

2: if $\text{StoppingCriteria}(D_L)$ then (2)

3: $n \rightarrow \text{left_prediction} = \text{FindPrediction}(D_L)$ (3)

4: else

5: **BuildSubtree** ($n \rightarrow \text{left}, D_L$)

6: if $\text{StoppingCriteria}(D_R)$ then

7: $n \rightarrow \text{right_prediction} = \text{FindPrediction}(D_R)$

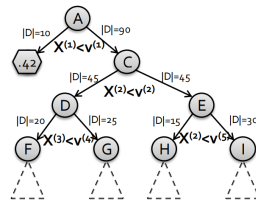
8: else

9: **BuildSubtree** ($n \rightarrow \text{right}, D_R$)

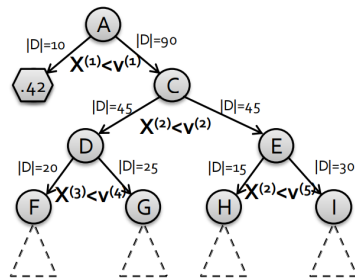
- ▶ (1) How to split? Pick attribute & value that optimizes some criterion
- ▶ Regression: Purity
 - Find split $(X^{(i)}, v)$ that creates D, D_L, D_R : parent, left, right child datasets and maximizes:

$$|D| \cdot \text{Var}(D) - (|D_L| \cdot \text{Var}(D_L) + |D_R| \cdot \text{Var}(D_R))$$

- $\text{Var}(D) = \frac{1}{|D|} \sum_{i \in D} (y_i - \bar{y})^2$... variance of y_i in D



- ▶ (1) How to split? Pick attribute & value that optimizes some criterion
- ▶ Classification: Information Gain
 - Measures how much a given attribute X tells us about the class Y
 - $IG(Y | X)$: We must transmit Y over a binary link. How many bits on average would it save us if both ends of the line knew X ?



The entropy of X :

$$H(X) = - \sum_{j=1}^m p(X_j) \log p(X_j)$$

- ▶ “High Entropy”: X is from a uniform (boring) distribution
 - A histogram of the frequency distribution of values of X is **flat**
- ▶ “Low Entropy”: X is from a varied (peaks/valleys) distribution
 - A histogram of the frequency distribution of values of X would have many lows and one or two highs



► Suppose I want to predict Y and I have input X

- X = College Major
- Y = Likes “Casablanca”

X	Y
Math	Yes
History	No
CS	Yes
Math	No
Math	No
CS	Yes
Math	Yes
History	No

► From this data we estimate

- $P(Y = \text{Yes}) = 0.5$

- $P(X = \text{Math} \wedge Y = \text{No}) = 0.25$
- $P(X = \text{Math}) = 0.5$
- $P(Y = \text{Yes} \mid X = \text{History}) = 0$

► Note:

- $H(Y) = -\frac{1}{2} \log_2(\frac{1}{2}) - \frac{1}{2} \log_2(\frac{1}{2}) = 1$
- $H(X) = 1.5$

- ▶ Suppose I want to predict Y and I have input X
 - $X = \text{College Major}$
 - $Y = \text{Likes "Casablanca"}$
- ▶ Def: Specific Conditional Entropy
 - $H(Y | X = v) =$ The entropy of Y among only those records in which X has value v
- ▶ Example:
 - $H(Y | X = \text{Math}) = 1$
 - $H(Y | X = \text{History}) = 0$
 - $H(Y | X = \text{CS}) = 0$

- ▶ Suppose I want to predict Y and I have input X
 - X = College Major
 - Y = Likes “Casablanca”
- ▶ Def: Conditional Entropy
 - $H(Y | X)$ = The average specific conditional entropy of Y
 - = the entropy of Y , conditioned on X , if you choose a record at random
 - = $\sum_j P(X = v_j)H(Y | X = v_j)$

- ▶ Suppose I want to predict Y and I have input X
 - X = College Major
 - Y = Likes “Casablanca”
- ▶ $H(Y | X)$ = The average specific conditional entropy of Y

$$= \sum_j P(X = v_j) H(Y | X = v_j)$$

- ▶ Example:

v_j	$P(X = v_j)$	$H(Y X = v_j)$
Math	0.5	1
History	0.25	0
CS	0.25	0

- ▶ So: $H(Y | X) = 0.5 \cdot 1 + 0.25 \cdot 0 + 0.25 \cdot 0 = 0.5$

► Suppose I want to predict Y and I have input X

- $X = \text{College Major}$
- $Y = \text{Likes "Casablanca"}$

► Def: Information Gain

- $IG(Y | X) =$ I must predict Y . How much information do I get about Y if I knew X ?

$$IG(Y | X) = H(Y) - H(Y | X)$$

► Example:

- $H(Y) = 1$
- $H(Y | X) = 0.5$
- Thus $IG(Y | X) = 1 - 0.5 = 0.5$

- ▶ Suppose you are trying to predict whether someone is going to live past 80 years
- ▶ From historical data you might find:
 - $IG(LongLife \mid HairColor) = 0.01$
 - $IG(LongLife \mid Smoker) = 0.4$
 - $IG(LongLife \mid Gender) = 0.25$
 - $IG(LongLife \mid LastDigitOfSSN) = 0.00001$
- ▶ IG tells us how much information about Y is contained in X
 - So attribute X that has high $IG(Y \mid X)$ is a good split!

3 steps in constructing a tree

Algorithm 1 **BuildSubtree**

Require: Node n , Data $D \subseteq D^*$

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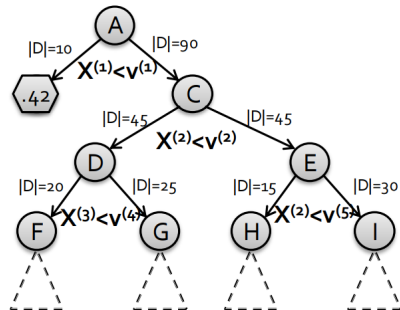
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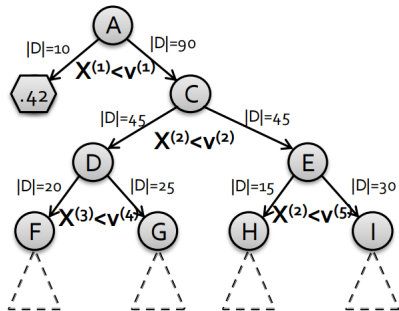
8: else

9: **BuildSubtree** ($n \rightarrow \text{right}, D_R$)

- ▶ (2) When to stop?
- ▶ Many different heuristic options
- ▶ **Two ideas:**
 - (1) When the leaf is “pure”
 - ▶ The target variable does not vary too much:
 $\text{Var}(y) < \varepsilon$
 - (2) When # of examples in the leaf is too small
 - ▶ For example, $|D| \leq 100$



- ▶ (3) How to predict?
- ▶ Many options
 - Regression:
 - ▶ Predict average y_i of the examples in the leaf
 - ▶ Build a linear regression model on the examples in the leaf
 - Classification:
 - ▶ Predict most common y_i of the examples in the leaf



► Characteristics

- Classification & Regression
 - Multiple (~ 10) classes
- Real valued and categorical features
- Few (hundreds) of features
- Usually dense features
- Complicated decision boundaries
 - Early stopping to avoid overfitting!

► Example applications

- User profile classification
- Landing page bounce prediction

► **Decision trees are the single most popular data mining tool:**

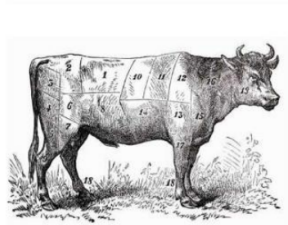
- Interpretable
- Easy to implement
- Easy to use
- Computationally cheap
- It's possible to mitigate overfitting (i.e., with ensemble methods)
- **They do classification as well as regression!**

► Learn multiple trees and combine their predictions

- Fix overfitting/underfitting problem in decision trees
- Gives better performance in practice
- The “wisdom of the crowds”

► The parable of the ox (Sir Francis Galton, 1906)

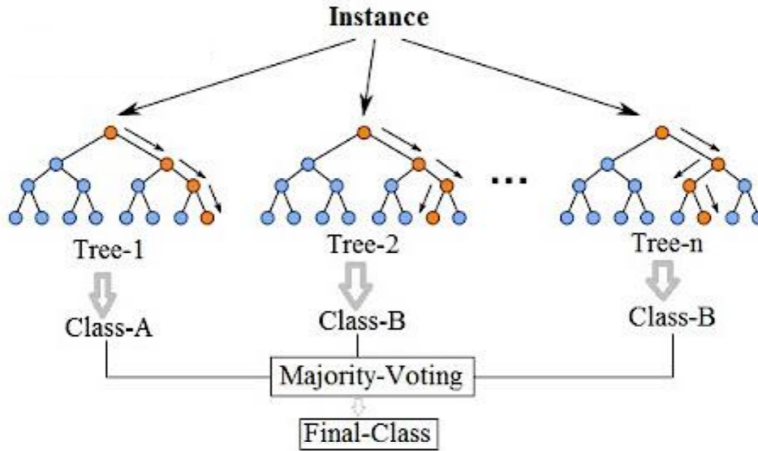
- 787 people guessed the weight of an ox
- Avg crowd guess: 1,197 pounds
- True weight: 1,198 pounds



Bagging (bootstrap aggregation):

- ▶ Learns multiple trees in parallel over independent samples of the training data
- ▶ **1) Bootstrapping:** Given a dataset D on n data points: Create multiple datasets D' of n points by sampling from D with replacement:
 - 37% points in D' will be duplicates, 63% will be unique
- ▶ **2) Parallel training:** Train decision trees on samples independently and in parallel
- ▶ **3) Aggregation:** Depending on the task, an average or majority of the predictions are computed for a more accurate estimate

(1): Bagging Decision Trees



(1): Instance Bagging

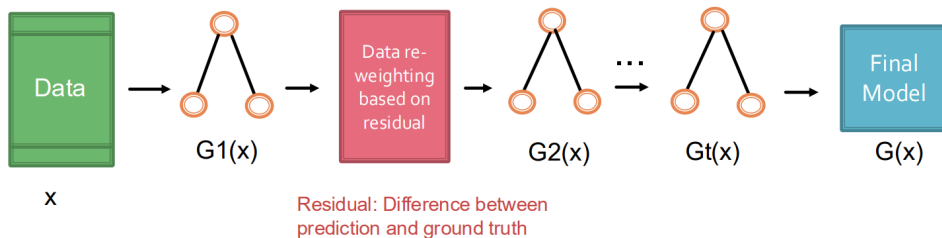
- ▶ Decision trees are greedy
 - They choose which variable to split on using a greedy algorithm that maximizes purity or information gain
- ▶ Even with Bagging, the decision trees can have a lot of structural similarities and correlation in their predictions
 - If one feature is very strong predictor, then every tree will select it, causing trees to be correlated.
- ▶ But ensemble learning works best with independent predictors

(2) Improvement: Random Forests

- ▶ Train a Bagged Decision Tree
- ▶ But use a modified tree learning algorithm that selects (at each candidate split) a random subset of features
 - If we have d features, consider $\sqrt{d}$ random features
- ▶ This is called: Feature bagging
 - Benefit: Breaks correlation between trees
- ▶ Random Forests achieve state-of-the-art results in many classification problems!

► **Boosting:** Another ensemble learning algorithm

- Combines the outputs of many “weak” classifiers to produce a powerful “committee”
- Learns multiple trees sequentially, each trying to improve upon its predecessor
- Final classifier is weighted sum of the individual classifiers



(3): Boosting

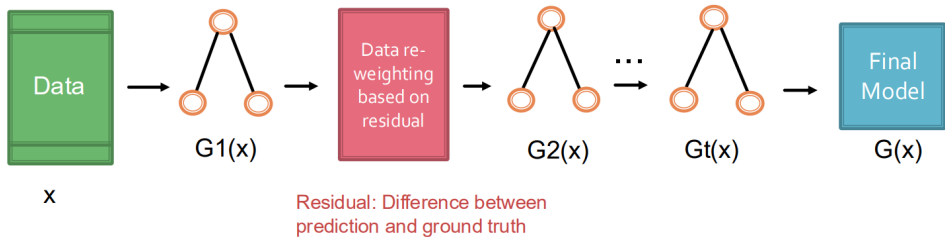
► We will show 2 examples:

- **Example 1: AdaBoost**

- Where each $G_t(x)$ is a one-level decision tree

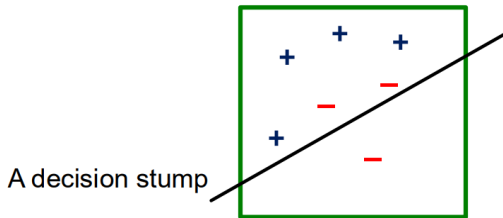
- **Example 2: Gradient Boosted Decision Trees**

- Where each $G_t(x)$ is a multi-level decision tree



► Decision “stumps”:

- 1-level decision tree
- A decision boundary based on one feature
 - E.g.: If someone is not a smoker, then predict them to live past 80 years old
- Building blocks of AdaBoost algorithm
- **Decision stump** is a **weak learner**



Boosting theory:
if weak learners have
>50% accuracy then
we can learn a perfect
classifier.

Suppose we have training data

$$\{(x_i, y_i)\}_{i=1}^N, \quad y_i \in \{1, -1\}$$

- ▶ Initialize equal weights for all observations: $w_i = \frac{1}{N}$
- ▶ At each iteration t :
 1. Train a stump G_t using data weighted by w_i
 2. Compute the **misclassification error** adjusted by w_i
 3. Compute the weight of the current tree α_t
 4. Reweight each observation based on prediction accuracy

- Calculate the weighted misclassification error:

$$\text{err}_t = \frac{\sum_{i=1}^N w_i I(y_i \neq G_t(x_i))}{\sum_{i=1}^N w_i}$$

- Use the error score to weight the current tree in the final classifier:

$$\alpha_t = \log \left(\frac{1 - \text{err}_t}{\text{err}_t} \right)$$

A classifier with 50% accuracy is given a weight of zero;

- Use misclassification error and tree weight to reweight the training data:

$$w_i \leftarrow w_i \exp [\alpha_t I(y_i \neq G_t(x_i))]$$

Training instances that are harder to classify get higher weight

- ▶ Final prediction is a weighted sum of the predictions from each stump:

$$G(x) = \text{sign} \left[\sum_{t=1}^T \alpha_t G_t(x) \right]$$

- ▶ More accurate trees are weighted higher in the final model

1. Initialize the observation weights $w_i = \frac{1}{N}$, $i = 1, 2, \dots, N$.
2. For $m = 1$ to M :

2.1 Fit a classifier $G_m(x)$ to the training data using weights w_i .

2.2 Compute

$$\text{err}_m = \frac{\sum_{i=1}^N w_i \mathbb{I}(y_i \neq G_m(x_i))}{\sum_{i=1}^N w_i}$$

2.3 Compute $\alpha_m = \log \left(\frac{1 - \text{err}_m}{\text{err}_m} \right)$.

2.4 Set $w_i \leftarrow w_i \cdot \exp [\alpha_m \cdot \mathbb{I}(y_i \neq G_m(x_i))]$, $i = 1, 2, \dots, N$.

3. Output

$$G(x) = \text{sign} \left[\sum_{m=1}^M \alpha_m G_m(x) \right]$$

- ▶ **Iteratively train weak learners (decision stumps) to form a strong model:**
 - Trees with high accuracy are given more weights in the final model
 - Misclassified data get higher weights in the next iteration
- ▶ AdaBoost is the equivalent to additive training with the exponential loss (Friedman et al. 2000)
- ▶ We will talk about additive training in more general scenarios next!

- ▶ Idea: Optimize an Additive model
- ▶ Additive prediction model:

$$\hat{y}_i = \sum_{t=1}^T f_t(x_i)$$

- Here f_t can be multi-level!
- ▶ Objective (cost) function:

$$\text{obj}(\theta) = \sum_{i=1}^N l(y_i, \hat{y}_i) + \sum_{t=1}^T \omega(f_t)$$

- $\omega(f_t)$ is a regularization term that models the complexity of the tree.

- Use Additive model to train sequentially:
 - Start from constant prediction, add a new decision tree f_i each time:

$$\hat{y}_i^{(0)} = 0$$

$$\hat{y}_i^{(1)} = f_1(x_i) = \hat{y}_i^{(0)} + f_1(x_i)$$

$$\hat{y}_i^{(2)} = f_1(x_i) + f_2(x_i) = \hat{y}_i^{(1)} + f_2(x_i)$$

...

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i)$$

- ▶ Prediction at round t is:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i)$$

- Where we need to decide what $f_t()$ to add
- ▶ Goal: Find tree $f_t(\cdot)$ that minimizes loss $l()$:

$$\text{obj}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t)}) + \omega(f_t)$$

- ▶ y_i : The ground-truth label
- ▶ $\hat{y}_i^{(t-1)} + f_t(x_i)$: The prediction made at round t
- ▶ $\omega(f_t)$: The model complexity

$$\text{obj}^{(t)} = \sum_{i=1}^n l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \omega(f_t)$$

- Take Taylor expansion of the objective:

$$g(x + \Delta) \approx g(x) + g'(x)\Delta + \frac{1}{2}g''(x)\Delta^2$$

- So, we get the approximate objective:

$$\text{obj}^{(t)} = \sum_{i=1}^n \left[l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \omega(f_t)$$

- where:

$$g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}_i^{(t-1)}), \quad h_i = \partial_{\hat{y}^{(t-1)}}^2 l(y_i, \hat{y}_i^{(t-1)})$$

- ▶ Our new goal: Find tree f_t that:

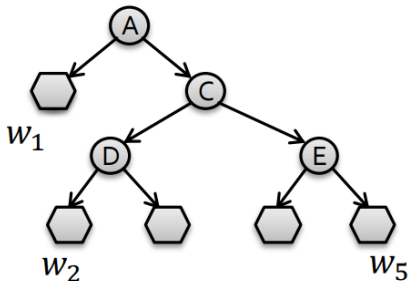
$$\sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \omega(f_t)$$

- ▶ Why spend so much effort to derive the objective, why not just grow trees...
 - **Theoretical benefit:** Know what we are learning
 - **Engineering benefit:**
 - ▶ g and h come from definition of loss function
 - ▶ Learning f_t only depends on the objective via g and h
 - ▶ We can now directly learn trees that optimize the loss (rather than using some heuristic procedure)

- ▶ Every leaf j has a weight w_j
 - We will predict w_j for any data belonging to leaf j

$$f_t(x) = w_{q(x)}$$

where $q(x)$ indicates the leaf node that data point x belongs to.



- Define complexity of tree f as:

$$\Omega(f) = \gamma \cdot T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

where:

- T : number of leaves of tree f
- γ : cost of adding a leaf to the tree f

► **Define:**

- The set of examples in the leaf j :

$$I_j = \{i \mid q(x_i) = j\}$$

where $q(x)$ denotes the leaf that data point x belongs to.

► The parameters that depend on the loss:

$$G_j = \sum_{i \in I_j} g_i \quad H_j = \sum_{i \in I_j} h_i$$

► Then the objective function becomes:

$$\text{obj}^{(t)} = \sum_{j=1}^T \left[G_j w_j + \frac{1}{2} (H_j + \lambda) w_j^2 \right] + \gamma T$$

- ▶ Given a tree f_t , we know how to

- Calculate the score for f :

$$\text{Obj} = -\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{H_j + \lambda} + \gamma T$$

- And then set optimal weights for the chosen f :

$$w_j^* = -\frac{G_j}{H_j + \lambda}$$

- ▶ In principle we could:

- Enumerate possible tree structures f and take the one that minimizes Obj

- ▶ In practice we grow tree greedily:
 - Start with tree with depth 0
 - For each leaf node in the tree, try to add a split
 - The change of the objective after adding a split is:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma$$

- Take the split that gives best gain
- ▶ Next: **How to find the best split?**

- ▶ For each node, enumerate over all features:
 - For each feature, sort the instances by feature value
 - Use a linear scan to decide the best split along that feature
 - Take the best split solution along all the features
- ▶ Pre-stopping:
 - Stop split if the best split has negative gain
 - But maybe a split can benefit future splits
- ▶ Post-pruning:
 - Grow a tree to maximum depth, recursively prune all the leaf splits with negative gain

- ▶ Add a new tree $f_t(x)$ in each iteration
 - Compute necessary statistics for our objective

$$g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)}), \quad h_i = \partial_{\hat{y}^{(t-1)}}^2 l(y_i, \hat{y}^{(t-1)})$$

- Greedily grow the tree that minimizes the objective:

$$Obj = -\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{H_j + \lambda} + \gamma T$$

- ▶ Add $f_t(x)$ to our ensemble model

$$y^{(t)} = y^{(t-1)} + \epsilon f_t(x_i)$$

ϵ is called step-size or shrinkage, usually set around 0.1. Goal: prevent overfitting.

- ▶ Repeat until we use M ensemble of trees

► XGBoost: eXtreme Gradient Boosting

- A highly scalable implementation of gradient boosted decision trees with regularization
- Widely used by data scientists and provides state-of-the-art results on many problems!
- System optimizations:
 - Parallel tree constructions using column block structure
 - Distributed Computing for training very large models using a cluster of machines.
 - Out-of-Core Computing for very large datasets that don't fit into memory.

- ▶ Basics of supervised learning
- ▶ **Decision Trees**
 - **Key idea:** split data at each internal node, make prediction at each leaf node
 - **How to construct a tree:** Information Gain
- ▶ **Ensemble of decision trees:**
 - **Bagging:** Random forests
 - **Boosting:** Boosted decision trees

- [1] Jure Leskovec, Anand Rajaraman, Jeff Ullman. *Mining of Massive Datasets*. MMDS 2020.
<http://www.mmds.org/>.